

Problem Set 2 – Interest Rate Parity

1. Suppose that the treasurer of IBM has an extra cash reserve of \$100,000,000 to invest for 12 months. The 12-month interest rate is 4 percent in the United States and 3.5 percent in Germany. Currently, the spot exchange rate is €1.01 per dollar and the 12-month forward exchange rate is €0.99 per dollar.

- a) Ignoring transaction costs, is covered interest rate parity holding (YES/NO/UNCERTAIN)?
- b) Ignoring transaction costs, is uncovered interest rate parity holding (YES/NO/UNCERTAIN)?
- c) In which country should the treasurer of IBM invest to maximize the return?

2. Currently, the spot exchange rate is \$1.50/£ and the 12-month forward exchange rate is \$1.52/£. The 12-month interest rate is 2.0% in the U.S. and 1.45% in the U.K. Assume that you can borrow as much as \$1,500,000 or £1,000,000.

- a. Since IRP is not holding, in which country would you borrow and in which country would you invest to make arbitrage profits?
- b. Show all the steps and determine the arbitrage profit for a U.S. investor (i.e., increase profit in dollar terms)

3. Suppose you see in the paper that 1-year interest rates are 5.93% in the United States and 70.0% in Turkey. Based on interest rate parity –is this possible in the absence of default risk? Can you make a prediction about what you expect to happen to the Turkish lira (versus the U.S. dollar) over the coming year?